Automated Irregular Shaped Stone Stacking via Pose Optimization and Monte Carlo Tree Search

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***Abstract*—Stacking irregular objects into prescribed target shapes without container walls requires mutually supporting arrangements despite uneven contact surfaces. The problem couples combinatorial object selection and ordering with continuous pose optimization, as each placement redistributes loads and changes support geometry and space available for subsequent objects. We propose a framework with two complementary components: structure-aware bilevel pose optimization and parallel Monte Carlo Tree Search (MCTS). The optimizer combines target-shape objectives with contact-force optimization to account for force and torque equilibrium across the structure under friction constraints, using bodywise ADMM updates for GPU parallelization. The receding-horizon MCTS planner evaluates batches of candidate placements in parallel through pose refinement and physical simulation, providing lookahead over object choices and placement poses. We hypothesize that combining structural stability evaluation with lookahead improves target coverage while maintaining stability under a fixed planning budget.**



Fig. 1: Preliminary stacking results. Left: simulated structures for different target shapes. Right: a physical stone structure from an outdoor experiment.

I. Introduction

Stacking is a fundamental problem with applications in bin packing, industrial warehouse storage [20, 9], and stone-wall construction [12]. In freestanding stacking, the resulting structure must approximate a prescribed target shape while maintaining force and torque equilibrium under contact forces and gravity, without support from an enclosing container. For n irregular objects, this geometric goal can be expressed as minimizing the unfilled volume of a target shape $T \subset \mathbb{R}^3$, subject to physical feasibility:

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathbb{V}\left(T \setminus \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \tag{1}$$
$$c_{\text{phys}}(O_1(q_1),\cdots,O_n(q_n)) = 0,$$

where $O_i(q_i) \subset \mathbb{R}^3$ is the region occupied by object i at pose $q_i \in \mathrm{SE}(3)$, $\mathbb{V}(\cdot)$ denotes volume, and $c_{\text{phys}} = 0$ represents stability and non-interpenetration constraints. Achieving these requirements simultaneously is challenging because changing one object pose alters both the occupied space and the contact geometry through which loads are transmitted across the structure. Moreover, a placement that fits well and is stable at the current stage may limit how closely the completed structure can approximate the target shape. Planning its construction therefore involves combinatorial choices of objects and placement order, together with iterative numerical evaluation of contact interactions [7], making planning computationally expensive.

To solve this problem, we parallelize

Existing automated stone-stacking methods use geometric criteria tailored to tower building [7] or wall construction [12, 17]. Extensions to freeform walls [13] broaden the range of wall geometries, but retain placement criteria specific to wall construction. Our geometric objectives instead use the prescribed target shape directly, allowing the same pose optimization formulation to be applied to different target geometries.

For structural stability, Liu and Napp [16] evaluate structural strength through simulation, using candidate generation and filtering rather than continuous pose refinement driven by structural equilibrium. Our optimizer uses equilibrium across the entire structure to guide placement adjustments before simulation. For sequential planning, Menezes et al. [18] learn placement values that account for future rewards, but assume a prescribed object sequence and fixed orientations. Our planner searches over both object choices and placement poses to account for their effects on subsequent placements.

II. Related Works

A. Automated Stone Stacking

Prior stone-stacking methods combine pose search with simulation [7] or use geometric placement rules for wall construction [12, 13, 17]. Liu and Napp [16] incorporate structural strength into sequence planning. Menezes et al. [18] learn placement values with a fixed object sequence and fixed orientations, and their elevation-map strategy informs our initial pose sampler. We combine target-driven pose optimization and structural equilibrium evaluation with explicit lookahead over object choices and placement poses.

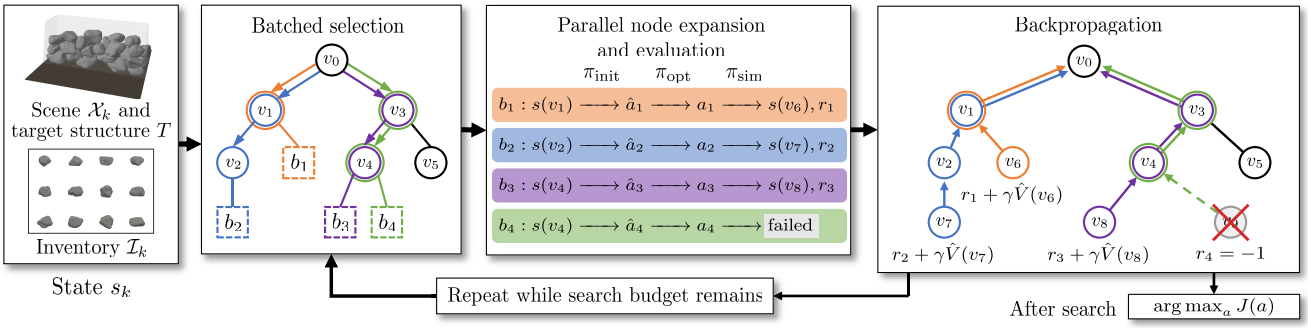


Fig. 2: Batched MCTS sequence planning. Colored requests are evaluated in parallel. Returns combine rewards and discounted leaf estimates. The dashed branch backpropagates its failure penalty once ($r_4 = -1$, $\hat{V}(v_9) = 0$) and is never selected again. After search, $J(a)$ ranks eligible root actions. See Sec. V-D for $\hat{V}(v)$ and $J(a)$.

B. Contact-Aware Pose Optimization

BODex [5] couples lower-level contact-force optimization with upper-level grasp synthesis, while GraspQP [21] defines a differentiable force-closure energy through a quadratic program. Our optimizer couples candidate placement with structure-wide equilibrium, using bodywise force decomposition for parallel evaluation.

C. Simulation-Based Sequential Manipulation Planning

Parallel Monte Carlo Tree Search with Batched Simulations (PMBS) [8] couples parallel search with GPU rigid-body simulation for object retrieval from clutter. Our stacking planner adopts batched evaluation, integrating pose refinement and structural simulation at each expansion.

III. Preliminaries and Overview

A. Shape Representation

We represent the target shape T and each object O_i as unions of smooth convex bodies modeled by differentiable support functions (DSFs) [14]. For a compact convex body $C \subset \mathbb{R}^3$, its support function is $h_C(x) = \max_{p \in C} x^\top p$ for $x \neq 0$. At pose $q = (R, t) \in \mathrm{SE}(3)$, we write

$$C(q) = \{Rp + t \mid p \in C\}, \qquad O_i(q_i) = \bigcup_j C_{ij}(q_i),$$

where C_{ij} are the convex components in object i's local frame. Similarly, $T = \bigcup_j T_j$, where T_j are fixed convex components in the world frame. DSFs support differentiable signed-distance and contact-geometry computations between posed convex bodies [2], enabling gradient-based pose optimization.

B. Markov Decision Process Formulation

Given an inventory of n objects, we model stacking as a finite-horizon sequential decision process in which one object is placed at each step. Let $\mathcal{P}_k \subseteq \{1, \dots, n\}$ be the indices of the objects already placed, and let $i_0 \notin \{1, \dots, n\}$ index the fixed ground. The current scene is $\mathcal{X}_k = \{(i, q_i) \mid i \in \mathcal{P}_k\}$, and the remaining inventory is $\mathcal{I}_k = \{1, \dots, n\} \setminus \mathcal{P}_k$. The state, action, and deterministic transition function are

$$s_k = (\mathcal{X}_k, \mathcal{I}_k, T), \quad a_k = (i_{k+1}, q_{k+1}), \quad s_{k+1} = F(s_k, a_k),$$

where $i_{k+1} \in \mathcal{I}_k$. The transition inserts the proposed pose into the scene, simulates the resulting structure, and returns the settled scene with $\mathcal{I}_{k+1} = \mathcal{I}_k \setminus \{i_{k+1}\}$. The reward $r(s_k, a_k)$ scores target-contained volume and physical quality; feasibility checks separately reject invalid transitions (Sec. V). The sequence planner therefore operates on the MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, F, r, \gamma)$.

C. Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) incrementally constructs a search tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ over the MDP. Each node $v \in \mathcal{V}$ represents a state $s(v)$, and each edge $(v, a) \in \mathcal{E}$ represents an action and its transition to a child node. Let $N(v)$ denote the number of completed evaluations and $Q(v)$ the empirical mean of discounted returns backpropagated to that node.

Each iteration selects a path by balancing value and exploration, expands and evaluates an action, and backpropagates its discounted return to update Q and N.

In our planner (Fig. 2), expansion uses three modules: π_{init} proposes an object and initial pose $\hat{a}_k = (i_{k+1}, \hat{q}_{k+1})$, π_{opt} refines the pose, and π_{sim} evaluates the resulting structure to determine feasibility, the settled successor, and reward. The resulting flow is

$$s_k \xrightarrow{\pi_{\text{init}}} \hat{a}_k \xrightarrow{\pi_{\text{opt}}} a_k \xrightarrow{\pi_{\text{sim}}} s_{k+1}, \; r(s_k, a_k).$$

Sec. V describes how these modules are integrated into batched search.

IV. Pose Optimization

The module π_{opt} refines $\hat{q}_{k+1}$ from π_{init} toward a stable, physically feasible placement q_{k+1}.

A. Single Object Pose Optimization Problem

Given s_k, we initialize q at $\hat{q}_{k+1}$ and solve the bilevel problem

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \leq 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

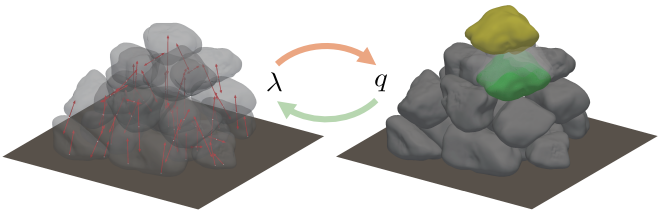
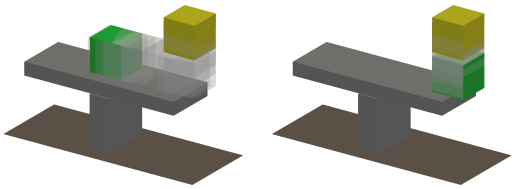


Fig. 3: Bilevel pose optimization. Left: optimizing contact forces λ under friction constraints minimizes structural equilibrium residuals at fixed q. Right: refining $q \in \mathrm{SE}(3)$ minimizes the geometric and optimized physical objectives.



(a) Structure-wide (b) Candidate-only

Fig. 4: Seesaw example: initial (yellow), optimized (green), and intermediate (gray) poses. Structure-wide equilibrium (a) moves the candidate toward the support; candidate-only equilibrium (b) leaves it near the unsupported end.

where $q \in \mathrm{SE}(3)$ is the candidate object's pose and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ collects all contact forces in the resulting structure, including inter-object and ground contacts. This formulation follows prior contact-aware optimization methods [15, 5].

The geometric objective f_{geo} is designed as follows:

$$f_{\mathrm{geo}}(q) = w_{\mathrm{h}} f_{\mathrm{h}}(q) + w_{\mathrm{in}} f_{\mathrm{in}}(q \mid T, O_{i_{k+1}}) + w_{\mathrm{pen}} f_{\mathrm{pen}}(q \mid \mathcal{X}_k, O_{i_{k+1}})$$

with positive weights. For $q = (R, t)$, $f_{\mathrm{h}}(q) = t_z$ is the world-frame height of the candidate's pose origin. Writing $O = O_{i_{k+1}}$, we penalize proximity to existing objects and the fixed ground as

$$f_{\mathrm{pen}}(q) = \sum_{j \in \mathcal{P}_k \cup \{i_0\}} \left[\epsilon_{\mathrm{margin}} - \mathrm{d}(O(q), O_j(q_j))\right]_+^2,$$

where d is signed distance (positive for separation), $[x]_+ = \max(x, 0)$, and $\epsilon_{\mathrm{margin}} \geq 0$ is the clearance margin. With $h_{O(q)}(u) = \max_j h_{C_{i_{k+1},j}(q)}(u)$, we define

$$m_{\mathrm{in}}(q) = \max_j \min_{u \in \mathbb{S}^2} \left[h_{T_j}(u) - h_{O(q)}(u)\right],$$
$$f_{\mathrm{in}}(q) = \left[-m_{\mathrm{in}}(q)\right]_+^2.$$

This term penalizes violations of containment of the entire object within a single convex target component, a conservative condition for containment in T. Noting that the term is differentiable almost everywhere in q, we compute pose derivatives at regular inner solutions by implicit differentiation. Nonsmooth transitions have not caused practical difficulties in our implementation.

Let $\mathcal{P}_{k+1} := \mathcal{P}_k \cup \{i_{k+1}\}$. For distinct $l, m \in \mathcal{P}_{k+1} \cup \{i_0\}$, let $I_{l,m} \subset \{1, \ldots, n_c\}$ index contacts between bodies l and m. Each object's contact set $I_l := \bigcup_{m \in (\mathcal{P}_{k+1} \cup \{i_0\}) \setminus \{l\}} I_{l,m}$ includes ground contacts, but the physical objective excludes the ground's equilibrium residual:

$$f_{\mathrm{phys}}(q, \lambda) = w_{\mathrm{eq}} \sum_{l \in \mathcal{P}_{k+1}} \|J_{c,I_l}^{\top}(q)\lambda_{I_l} + \lambda_{\mathrm{ext},l}\|_W^2 + w_{\mathrm{cp}} \sum_{j \in I_{i_{k+1}}} \mathrm{d}_j(q)^2 \|\lambda_j\|^2 + w_{\mathrm{reg}} \sum_{j=1}^{n_c} \|\lambda_j\|^2 \quad (3)$$

The three terms respectively penalize violations of static equilibrium, contact complementarity between the candidate and the current scene, and large contact forces through regularization. Figure 4 illustrates why equilibrium is evaluated across the structure rather than only for the candidate. The weight matrix $W \succ 0$ balances force and moment scales. Here, $\mathrm{d}_j(q)$ is the signed distance between the bodies of contact candidate j; J_{c,I_l}, $\lambda_{\mathrm{ext},l}$, and λ_{I_l} are object O_l's contact Jacobian, external wrench, and contact-force vector, respectively; and $w_{\mathrm{eq}}, w_{\mathrm{cp}}, w_{\mathrm{reg}} > 0$ are objective weights.

Finally, each $g_{\mathrm{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\mathrm{cone},j}(q, \lambda_j) = \sqrt{\|J_{c,j}^t \lambda_j\|^2 + \epsilon} - \mu_j J_{c,j}^n \lambda_j \quad (4)$$

where $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J_{c,j}^t$ and $J_{c,j}^n$ are the tangential and normal components of the Jacobian of the contact constraint.

B. Upper-Level Problem Optimization on SE(3) Manifold

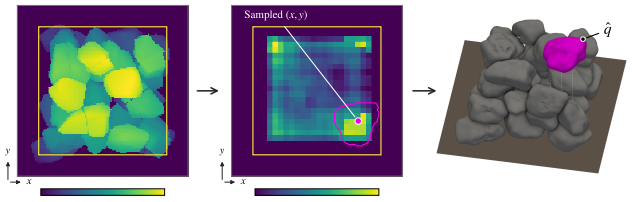
Let $\lambda^*(q)$ be the solution of the lower-level problem and define $f_{\mathrm{phys}}(q) := f_{\mathrm{phys}}(q, \lambda^*(q))$. The envelope theorem [4] gives

$$\frac{df_{\mathrm{phys}}(q)}{dq} = \frac{\partial f_{\mathrm{phys}}(q, \lambda^*)}{\partial q} + \sum_{j=1}^{n_c} \nu_j^* \frac{\partial g_{\mathrm{cone},j}(q, \lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable for the j-th friction cone constraint. Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with BFGS Hessian approximation [10]. A homotopy schedule adjusts the geometric and physical weights between stages, holding them fixed within each stage and initializing from the preceding pose. It progressively reduces w_{reg} to limit regularization-induced equilibrium residuals. Candidate evaluation uses the final pose and forces.

C. Lower-Level Problem Optimization with ADMM

For fixed q, the lower-level problem is convex in λ. To apply bodywise ADMM [3], we introduce two local copies $\lambda_{j,1}$ and $\lambda_{j,2}$ per contact, one for each contacting body, including the ground. Collecting these copies into $\tilde{\lambda}$ makes each body's equilibrium residual depend only on its local variables.



(a) Scene height (b) Proposal score (c) Sampled initial pose

Fig. 5: Initial pose sampling for a fixed object and orientation: (a) scene height, (b) proposal score and sampled planar location, and (c) initial pose $\hat{q}$ above the scene. Magenta links the sampled location and candidate object.

Consensus and friction constraints give the reformulation

$$\begin{aligned} \min_{\tilde{\lambda},z} \quad & \tilde{f}_{\text{phys}}(q,\tilde{\lambda}) + \sum_{\substack{j=1,\dots,n_c \\ s=1,2}} \delta_{K_j}(z_{j,s}) \\ \text{s.t.} \quad & \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2}, \qquad j = 1,\dots,n_c. \end{aligned}$$

where $\tilde{f}_{\text{phys}}$ is the physical objective in local variables, $z_{j,s}$ are auxiliary variables, and δ_{K_j} is the indicator of $K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\text{cone},j}(q,\lambda) \le 0\}$. ADMM solves the body subproblems independently. An equivalent four-dimensional second-order cone (SOC) lifting enables closed-form cone projections for the friction updates, while consensus updates reduce to averaging.

The pose optimization module is implemented using CUDA to leverage GPU parallelization for efficient computation of the gradients and the ADMM updates.

V. Sequence Planning

A. Initial Pose Sampling

The sampler π_{init} uses volume-biased object sampling with a finite orientation set per object. It favors lower placements and seeds empty scenes near feasible target corners. For object i, orientation R, and feasible planar location $\mathbf{x}$, the score (Fig. 5) is

$$\sigma(\mathbf{x} \mid i, R) = w_{\text{s}}\sigma_{\text{s}} + w_{\text{f}}\sigma_{\text{f}} + w_{\text{b}}\sigma_{\text{b}} - w_{\text{h}}\sigma_{\text{h}} - w_{\text{d}}\sigma_{\text{d}},$$

with positive weights. The first four terms measure the fraction of bottom samples near predicted contact, unfilled layer area beneath the footprint, proximity to the feasible placement boundary, and normalized mean contact height, respectively. A bounded penalty σ_{d} on the planar footprint gap to the preceding object (zero for the first placement) encourages nearby placements, helping MCTS assess their interactions without excluding distant proposals. Inspired by Menezes et al. [18], the floor and boundary terms encourage filling empty layer regions and building inward from the boundary.

We combine elite selection with sampling without replacement weighted by $\exp(\sigma/\tau)$, $\tau > 0$. Proposals start above the scene with clearance and undergo π_{opt} refinement, including XY/yaw mutations at the root. Heuristic filters limit planar displacement from each initial pose after optimization and settling, and screen out candidates whose bottoms lie too far above the current construction level or whose equilibrium residual is too large (defined below).

B. Simulation Module

The module π_{sim} uses the same simulation procedure at every search depth. It settles the new object for a fixed duration with existing objects fixed, then resets velocities and simulates the entire structure's response to the added load. Structural simulation requires a minimum duration and sustained low mass-normalized kinetic energy. Excessive accumulated motion, invalid contacts, or failure to settle within the time limit rejects the placement.

The accepted settled scene defines the successor $s' = F(s,a)$. Each disturbance trial starts from rest in this scene and applies a horizontal force at the new object's center of mass:

$$\mathbf{F}_j = \alpha m_{i_{k+1}} g \big(\cos\theta_j,\ \sin\theta_j,\ 0\big)^\top,$$

where $\theta_j = 2\pi j/n_d$ for $j = 0,\dots,n_d-1$, $\alpha > 0$ sets the force relative to object weight, and g is gravitational acceleration. The force is removed after Δt_f to observe recovery; all placed objects remain dynamic throughout. Every trial must recover within prescribed tolerances without excessive pose excursion. These trials assess structural robustness without changing s'. GPU-accelerated subADMM [11] enables parallel candidate simulation, complementing the parallel pose optimizer for batched MCTS evaluation.

C. Reward Design

For the newly placed object $i = i_{k+1}$, let q_i' be its pose in the settled successor s' returned by π_{sim} (Sec. V-B). We define the target-volume and object-volume ratios

$$\rho_T = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(T)}, \quad \rho_O = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(O_i)}.$$

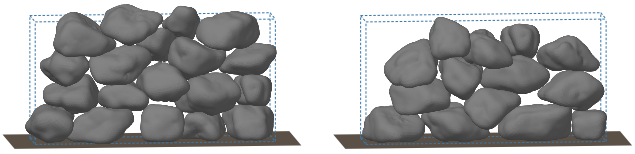
Here $\mathbb{V}(\cdot)$ denotes volume. The ρ_T term encourages target filling, while ρ_O discourages placements extending excessively outside the target. Intersection volumes are estimated on a voxel grid. Failed transitions receive $r(s,a) = -1$. For feasible transitions, with fixed normalization scales absorbed into positive weights, the reward is

$$\begin{aligned} r(s,a) = {} & w_T\rho_T + w_O[\rho_O - \tfrac{1}{2}]_+ - w_E e(a) \\ & + w_H\big(\phi_H(s) - \phi_H(s')\big) - w_D D, \end{aligned}$$

where $[x]_+ = \max(x,0)$, so the containment bonus is active above 50%. Coverage lost through motion of earlier objects is not subtracted. The height term rewards the decrease in surface-height variance $\phi_H(s) = \langle (H_s - \langle H_s\rangle_\Omega)^2\rangle_\Omega$ between parent and settled successor. Here H_s is height over the target footprint Ω, zero at empty cells, and $\langle\cdot\rangle_\Omega$ denotes the area-weighted mean.

The equilibrium score uses residual forces $\mathbf{f}_l$ and torques $\boldsymbol{\tau}_l$ returned by pose optimization:

$$e(a) = \max_l \left(\frac{\|\mathbf{f}_l\|}{m_l g} + \frac{\|\boldsymbol{\tau}_l\|}{m_l g \mathbb{V}(O_l)^{1/3}} \right).$$



(a) MCTS (b) Greedy

Fig. 6: Wall construction using the same inventory of over 40 stones. MCTS places 19 stones (44.0% target filling); greedy stops at 14 (36.1%), finding no further feasible placement. Blue dashed outlines show the target.

Here l indexes objects in the resulting structure and g is gravitational acceleration. Candidates with $e(a) \geq \epsilon_{\text{eq}}$ are rejected after pose optimization, before simulation, so the equilibrium term ranks only admitted placements. Let $q_l^{(j)}(t)$ denote object l's pose during force trial j (Sec. V-B), starting from its nominal successor pose q_l'. Disturbance sensitivity is

$$D = \min\left\{ D_{\max}, \max_{j,l,t} d_{\text{SE(3)}}\left(q_l^{(j)}(t), q_l'\right) \right\},$$

where $d_{\text{SE(3)}}$ sums Euclidean translation distance and angular distance weighted by $\lambda_\theta > 0$. The maximum spans force trials, all placed objects, and simulation time including recovery; $D_{\max}$ caps the penalty. Physical rejection criteria apply independently of the reward.

D. Monte Carlo Tree Search

We integrate these modules in batched, receding-horizon MCTS (Fig. 2), following the parallel evaluation approach of [8]. Figure 6 shows a recorded comparison with greedy placement.

Batched Selection. For an optimized action $a = (i, q)$, let ρ_O^{pre} denote the object-volume ratio evaluated at q, before simulation. Using the reward weights, the logit $\ell(a) = w_O \rho_O^{\text{pre}} - w_E e(a)$ defines $P(v_p, a)$ through a softmax over selectable children. Writing $N_v = N(v)$ and similarly for Q_v and $\hat{N}_v$, an evaluated child v of v_p is selected using PUCT [19]:

$$\text{PUCT}(v) = \frac{N_v Q_v}{N_v + \hat{N}_v} + cP(v_p, a)\frac{\sqrt{N_{v_p} + \hat{N}_{v_p}}}{1 + N_v + \hat{N}_v}, \qquad (5)$$

where $c > 0$ controls exploration, $\hat{N}$ counts temporary reservations, and the first term is zero when its denominator is zero. At non-root nodes, progressive widening [6] increases the number of candidate actions with completed visits. Temporary reservations prevent duplicate evaluations within a batch without increasing the widening allowance.

Parallel Expansion and Evaluation. After batch selection, fresh proposals pass through $\pi_{\text{init}} \to \pi_{\text{opt}} \to \pi_{\text{sim}}$. Independent candidates are processed in parallel; selection never descends through an action without an evaluated successor. Failed candidates are excluded from further selection.

Leaf Value Estimation and Backpropagation. We replace rollouts with a heuristic estimate of future rewards:

$$\hat{V}(v) = \begin{cases} 0, & \text{if failed or terminal,} \\ \hat{r}\sum_{j=0}^{d_v - 1}\gamma^j, & \text{otherwise,} \end{cases}$$

where $\hat{r}$ is a prescribed per-step reward estimate and d_v the bootstrap horizon, capped by remaining depth and inventory (zero at the depth limit). Backup starts from $r_v + \gamma\hat{V}(v)$, where r_v is the incoming action's reward, and recursively adds ancestor rewards with discount γ, updating Q, N, and clearing temporary reservations.

Action Selection and Tree Reuse. For each feasible root action a, let $G(a)$ be the maximum discounted observed return among its deepest evaluated feasible paths within the search horizon, including a's reward and excluding bootstrap values. Root-only paths are eligible. We select the action maximizing

$$J(a) = (1 - \beta)Q(v_a) + \beta G(a), \qquad \beta \in [0, 1].$$

The path-return term counterbalances failures accumulated during exploration, which can depress the mean even when a promising feasible path exists. If no root action is feasible, we return no placement.

After execution, we advance the root to the observed successor. Evaluated descendants are retained when this state and the planning context match those used to construct the subtree; otherwise, the affected subtree and its statistics are discarded.

Acknowledgments

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